

Meter Management System (Workforce Management)

Full Stack Assignment (.NET API + MVC)

Objective

Build a system to manage meters and track their status using .NET Web API as backend and MVC as frontend.

Architecture Requirements

- 1 Backend: .NET Web API (RESTful APIs).
- 2 Frontend: ASP.NET MVC consuming the APIs.
- 3 Use Entity Framework for database operations.
- 4 Apply layered architecture (Controller, Service, Repository).

Required Screens

- 1 Login
- 2 Sign Up
- 3 Add Meter (single or batch upload)
- 4 Assign Meter to Users (Admin only)

Core Features

- 1 User authentication (Admin / Agent roles).
- 2 Add meters individually or in bulk.
- 3 Assign meters to agents (Admin only).
- 4 Track meter status:
 - On Stock
 - Assigned to Agent
 - Installed at Customer
- 5 Automatically update meter status when assigned.

API Requirements

- 1 Authentication APIs (Login / Register).
- 2 CRUD APIs for Meters.
- 3 Assign Meter API.
- 4 Get meters by status or assigned user.
- 5 Use proper HTTP verbs (GET, POST, PUT,DELETE).

MVC Requirements

- 1 Consume
- 2 Use ViewModels.
- 3 Implement validation.
- 4 Display meter status clearly in UI

Important

- 1 Pagination & search.
- 2 Error handling & logging.
- 3 Role-based authorization.
- 4 Clean architecture (Repository pattern)

Evaluation Criteria

- 1 Correct use of Entity Framework.
- 2 Clean API design.
- 3 Separation between API and MVC.
- 4 Code quality and structure.
- 5 Handling edge cases.